BE-201 · BTech First Year · UIT, Barkatullah University, Bhopal

Engineering Physics — Short Notes

Unit-wise complete syllabus coverage · Hinglish notes · Prepared by Ajeet Soni

UNIT I Quantum Physics

1. Phase Velocity, Group Velocity aur unka Relation

de Broglie: har moving particle ke saath **matter wave**, $\lambda = h/p = h/mv$. Particle ko represent karne ke liye kai waves superpose karke **wave packet** banate hain.

PHASE VELOCITY Constant phase point ki speed: $v_p = \omega/k$. Matter wave: $v_p = E/p = c^2/v > c$ (no energy carry, physically meaningless).

GROUP VELOCITY Wave packet (envelope) ki speed: $v_g = d\omega/dk$. Matter wave: $v_g = v$ (particle velocity) — energy isi se travel.

- $y_1 = A\sin(\omega t - kx)$, $y_2 = A\sin[(\omega+\Delta\omega)t - (k+\Delta k)x]$ add karo ⇒ $y = 2A\cos(\Delta\omega t/2 - \Delta k x/2)\cdot\sin(\omega t - kx)$; envelope speed = $\Delta\omega/\Delta k \to d\omega/dk$.
- $E = \hbar\omega$, $p = \hbar k$ ⇒ $v_g = dE/dp$; $E = p^2/2m$ ⇒ $v_g = p/m = v$. Relation: $\omega = kv_p$ ⇒ $v_g = v_p + k\, dv_p/dk$.

FORMULA

$$v_g = v_p - \lambda\,(dv_p/d\lambda)\ ;\ v_p \cdot v_g = c^2\ (\text{relativistic particle})$$

Non-dispersive medium ($dv_p/d\lambda = 0$) ⇒ $v_g = v_p$; normal dispersion ⇒ $v_g < v_p$.

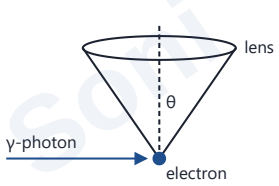
2. Heisenberg Uncertainty Principle

STATEMENT Particle ki position aur momentum **ek saath exactly** measure karna impossible: $\Delta x \cdot \Delta p_x \geq \hbar/2$ (elementary form $\Delta x\cdot\Delta p \approx h$). Isi tarah $\Delta E\cdot\Delta t \geq \hbar/2$, $\Delta L\cdot\Delta\theta \geq \hbar/2$.

Elementary proof: k aur k+Δk waves ke superposition se ek group ki length $\Delta x = 2\pi/\Delta k$. $p = \hbar k$ ⇒ $\Delta p = \hbar\Delta k$ ⇒ $\Delta x\cdot\Delta p = 2\pi\hbar = h$. Packet chhota (Δx kam) ⇒ Δp zyada.

(a) Gamma-ray microscope se electron ki position

- Resolving limit: $\Delta x = \lambda/(2\sin\theta)$ (θ = semi-vertical angle).
- Scattered photon (h/λ) cone 2θ mein kahin bhi ⇒ recoil $\Delta p_x = 2(h/\lambda)\sin\theta$.
- ⇒ $\Delta x\cdot\Delta p_x = h$. Chhota λ ⇒ Δx kam par Δp zyada.



Heisenberg microscope

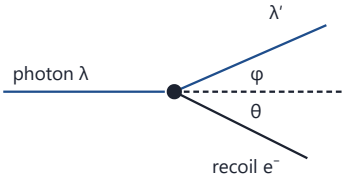
(b) Electron nucleus mein nahi: Nucleus dia ~ 2×10^{-14} m = Δx ⇒ $\Delta p \approx h/\Delta x \approx 3.3\times10^{-20}$ kg m/s ⇒ $E \approx pc \approx 10^{-11}$ J ≈ **60 MeV**. Par β-decay electrons sirf **2–4 MeV** ke hote hain ⇒ electron nucleus mein exist nahi kar sakta.

(c) Single slit se electron diffraction: Slit width d = Δy. First minimum $d\sin\theta = \lambda$. y-momentum $-p\sin\theta$ se $+p\sin\theta$ ⇒ $\Delta p_y = 2p\sin\theta = 2h/d$ ⇒ $\Delta y\cdot\Delta p_y = 2h \geq \hbar/2$. Narrow slit ⇒ zyada spread.

3. Compton Scattering

X-ray (λ) free electron se scatter hone par wavelength **badh jaati** hai (λ′ > λ) — light ke **photon (particle) nature** ka proof.

Derivation: Energy: $h\nu + m_0c^2 = h\nu' + mc^2$. Momentum: $h/\lambda = (h/\lambda')\cos\varphi + p\cos\theta$, $0 = (h/\lambda')\sin\varphi - p\sin\theta$. θ eliminate karo + $E^2 = p^2c^2 + m_0{}^2c^4$ ⇒



Compton geometry

COMPTON SHIFT

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_0 c}(1 - \cos\varphi)$$

h/m_0c = **0.02426 Å** (Compton wavelength). φ = 90° ⇒ 0.0243 Å; φ = 180° ⇒ $\Delta\lambda_{max}$ = 0.0485 Å. **Numerical:** λ = 1 Å, φ = 60° ⇒ Δλ = 0.0121 Å, λ′ = 1.0121 Å. Δλ λ pe depend nahi karta ⇒ visible light mein negligible.

4. Wave Function ψ, Properties & Operators

ψ(x,t) matter wave ka amplitude (complex, khud physical nahi); $|\psi|^2 = \psi^*\psi$ = **probability density** (Born).

- **Normalization:** $\int |\psi|^2 \, dV = 1$.
- ψ **finite, single-valued, continuous**.
- ∂ψ/∂x continuous (finite V pe).
- ψ → 0 as x → ±∞ (bound states).
- Orthogonal: $\int \psi_m^* \psi_n \, dx = 0$ (m ≠ n).
- Aisa ψ = **well-behaved** wave function.

OPERATORS

$$\hat{p} = -i\hbar\,\partial/\partial x \;|\; \hat{E} = i\hbar\,\partial/\partial t \;|\; \hat{H} = -(\hbar^2/2m)\nabla^2 + V \;|\; \langle A\rangle = \int \psi^* \hat{A} \psi \, dx$$

Origin: $\psi = A\, e^{i(kx-\omega t)}$ ⇒ $\partial\psi/\partial x = (ip/\hbar)\psi \Rightarrow p\psi = -i\hbar\partial\psi/\partial x$; $\partial\psi/\partial t = -(iE/\hbar)\psi \Rightarrow E\psi = i\hbar\partial\psi/\partial t$.

5. Schrödinger Wave Equation

TDSE $E = p^2/2m + V$; ψ se multiply karke operators daalo:

$$i\hbar \frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi + V\psi$$

TISE $V \neq V(t) \Rightarrow \psi(x,t) = \psi(x)e^{-iEt/\hbar}$ TDSE mein daalo:

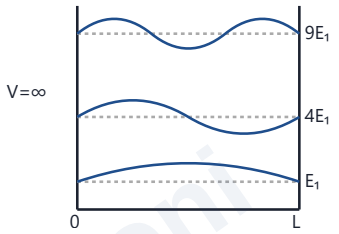
$$\nabla^2\psi + \frac{2m}{\hbar^2}(E - V)\psi = 0$$

Alternate TISE: wave eq. $\nabla^2\psi + (4\pi^2/\lambda^2)\psi = 0$ mein λ = h/p, $p^2 = 2m(E-V)$ substitute karo.

6. Particle in 1-D Infinite Square Potential Well

V = 0 (0 < x < L), V = ∞ bahar ⇒ bahar ψ = 0. Andar: $d^2\psi/dx^2 + k^2\psi = 0$, $k^2 = 2mE/\hbar^2$.

- $\psi = A\sin kx + B\cos kx$; ψ(0) = 0 ⇒ B = 0.
- ψ(L) = 0 ⇒ sin kL = 0 ⇒ $k = n\pi/L$, n = 1,2,3…
- $E_n = \hbar^2k^2/2m = n^2h^2/8mL^2$.
- Normalize: $A^2L/2 = 1 \Rightarrow A = \sqrt{(2/L)}$.



ψ_1, ψ_2, ψ_3 in the well

RESULT

$$E_n = \frac{n^2h^2}{8mL^2} \;;\; \psi_n(x) = \sqrt{(2/L)}\sin(n\pi x/L)$$

n = 0 nahi (ψ ≡ 0) ⇒ **zero-point energy** $E_1 \neq 0$. Energy quantized, $E \propto n^2$, $E \propto 1/L^2$. **Numerical:** electron, L = 1 Å ⇒ $E_1 = (6.63\times10^{-34})^2/(8\times9.1\times10^{-31}\times10^{-20}) = 6.03\times10^{-18}$ J = **37.7 eV**; E_2 = 150.8 eV.

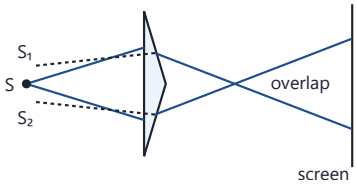
UNIT II Wave Optics

Interference: do **coherent** sources (same frequency, constant phase difference) ki waves ka superposition. Path diff. Δ = nλ ⇒ bright; Δ = (2n+1)λ/2 ⇒ dark; phase diff δ = (2π/λ)Δ.

1. Fresnel's Biprism

Do thin prisms (angle ~0.5°) base-to-base. Slit S se do **virtual coherent sources** S_1, S_2 (separation d) — **division of wavefront**.

- Fringe width $\beta = \lambda D/d \Rightarrow \lambda = \beta d/D$.
- d by displacement method: $d = \sqrt{(d_1 d_2)}$.
- Thin sheet (t, μ) ⇒ fringe shift $x_0 = (\mu-1)t\beta/\lambda$.



Fresnel biprism

2. Thin Films (Reflected & Transmitted Light)

Thickness t, index μ, refraction angle r: path diff. $\Delta = 2\mu t\cos r$; denser medium se reflection pe extra **λ/2** (Stokes).

Light	Bright	Dark
Reflected (extra λ/2)	2μt cos r = (2n+1)λ/2	2μt cos r = nλ
Transmitted (no λ/2)	2μt cos r = nλ	2μt cos r = (2n+1)λ/2

Dono patterns **complementary**. Very thin film (t → 0) reflected light mein dark. Soap/oil film white light mein coloured.

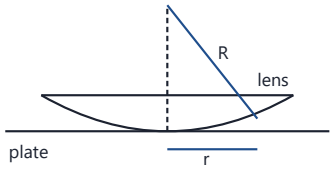
3. Wedge-shaped Film

Do plates ke beech chhota angle θ. Dark: $2\mu t = n\lambda$, $t = x\theta \Rightarrow$ straight, equidistant fringes, edge pe dark. $\beta = \lambda/2\mu\theta$; wire thickness $d = \theta l = \lambda l/2\mu\beta$.

4. Newton's Rings

Plano-convex lens (R) glass plate pe ⇒ air film; reflected light mein concentric rings, **centre dark**.

- Geometry: $r^2 \approx 2Rt$.
- Dark: $2\mu t = n\lambda \Rightarrow D_n^2 = 4n\lambda R/\mu$.
- Bright: $D_n^2 = 2(2n-1)\lambda R/\mu$. $D_n \propto \sqrt{n}$.



Newton's rings

FORMULA

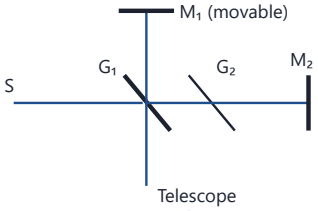
$$\lambda = \frac{D_{n+p}^2 - D_n^2}{4pR} \;;\; \mu_{liq} = \frac{(D_{n+p}^2 - D_n^2)_{air}}{(D_{n+p}^2 - D_n^2)_{liq}}$$

Numerical: D_5 = 0.336 cm, D_{15} = 0.590 cm, R = 100 cm ⇒ λ = (0.3481 − 0.1129)/(4×10×100) = **5880 Å**.

5. Michelson Interferometer

Beam splitter G_1 (half-silvered, 45°) do perpendicular beams banata hai → M_1 (movable), M_2 (fixed) se reflect → recombine. G_2 = **compensating plate** (dono beams ka glass path equal). M_1 ko d move karo ⇒ $2d = N\lambda$.

- **λ:** λ = 2d/N.
- **Δλ (Na D-lines):** successive concordance positions ke beech d ⇒ $\Delta\lambda = \lambda^2/2d$.
- **Thin sheet:** $2(\mu-1)t = N\lambda$. Also metre standardisation.



Michelson interferometer

6. Diffraction (Fraunhofer)

Single slit (width a)	Double slit (d = a+b)
Minima: $a\sin\theta = n\lambda$. $I = I_0(\sin\alpha/\alpha)^2$, $\alpha = \pi a \sin\theta/\lambda$. Secondary maxima α ≈ 1.43π, 2.46π (4.5%, 1.6%). Central max width 2λf/a.	$I = 4I_0(\sin\alpha/\alpha)^2\cos^2\beta$, $\beta = \pi(a+b)\sin\theta/\lambda$. Interference max: $(a+b)\sin\theta = n\lambda$; diffraction min: $a\sin\theta = m\lambda \Rightarrow$ **missing orders** $n = m(a+b)/a$ (a = b ⇒ 2, 4, 6… missing).

N SLITS — PLANE GRATING

$$I = I_0 \frac{\sin^2\alpha}{\alpha^2} \cdot \frac{\sin^2 N\beta}{\sin^2\beta} \;;\; \text{Principal maxima: } (a+b)\sin\theta = n\lambda$$

(a+b) = 1/(lines per unit length); $n_{max} < (a+b)/\lambda$. Principal maxima ke beech (N−1) minima, (N−2) secondary maxima; intensity ∝ N^2. **Numerical:** 5000 lines/cm, λ = 5890 Å ⇒ (a+b) = 2×10^{-4} cm, $\sin\theta_1 = 0.2945 \Rightarrow \theta_1 \approx 17.1°$; $n_{max} < 3.4 \Rightarrow$ 3rd order.

7. Resolving Power of Grating & Prism

Rayleigh criterion: ek wavelength ka principal max dusre ke first min pe pade to just resolved. RP = λ/dλ.

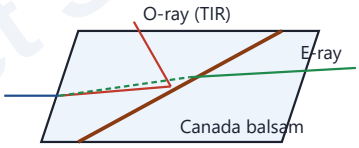
Grating: $\lambda/d\lambda = nN$	**Prism:** $\lambda/d\lambda = t\,(d\mu/d\lambda)$
nth max of (λ+dλ): $(a+b)\sin\theta = n(\lambda+d\lambda)$; first min after nth max of λ: $(a+b)\sin\theta = n\lambda + \lambda/N$. Equate ⇒ nN. Na D-lines (5890, 5896 Å), 1st order ⇒ N ≥ 5893/6 ≈ 982.	t = base length. Extreme rays ka path diff. $t\cdot d\mu = \lambda$ (Rayleigh). Cauchy $\mu = A + B/\lambda^2 \Rightarrow d\mu/d\lambda = -2B/\lambda^3$.

8. Polarized Light

Light **transverse** hai; plane polarized light mein E-vector ek hi plane mein. Malus: $I = I_0\cos^2\theta$. **Brewster's law:** polarizing angle i_p pe reflected light completely plane polarized: $\mu = \tan i_p$, aur $i_p + r = 90°$ (reflected ⊥ refracted). Glass $\mu = 1.5 \Rightarrow i_p \approx 56.3°$.

Double refraction: calcite/quartz mein ek ray do mein split: **O-ray** (Snell's law follow, μ_o constant) aur **E-ray** (μ_e direction-dependent); dono plane polarized, perpendicular planes. **Optic axis** ke along no double refraction. Calcite $\mu_o = 1.658$, $\mu_e = 1.486$ (negative crystal); quartz positive.

Nicol prism: calcite (end faces 71° → 68° cut) diagonally kaat kar **Canada balsam** ($\mu = 1.55$) se joda. O-ray (1.658 > 1.55) balsam pe **TIR** se side nikal jaati; E-ray (1.486 < 1.55) transmit ⇒ plane polarized output. Use: polarizer & analyser.



Nicol prism

Retarder (optic axis ∥ face)	Path diff.	Thickness	Use
Quarter wave plate	λ/4 (π/2)	$t = \lambda/4(\mu_o - \mu_e)$	Plane pol. at 45° → circular; other angles → elliptical
Half wave plate	λ/2 (π)	$t = \lambda/2(\mu_o - \mu_e)$	Plane of polarization 2θ rotate; circular ka sense reverse

Calcite, λ = 5890 Å ⇒ QWP $t = 5890\times10^{-10}/(4\times0.172) \approx 0.856$ μm. (Positive crystal: $\mu_e - \mu_o$.)

UNIT III Nuclear Physics

Nucleus: Z protons + N neutrons, A = Z + N; $R = R_0A^{1/3}$ ($R_0 \approx 1.2$ fm). Binding energy $B = [Zm_p + Nm_n - M]c^2$, 1 u = 931.5 MeV.

1. Liquid Drop Model & Semi-empirical Mass Formula

Nucleus = charged incompressible liquid drop. Similarities: short-range saturated force (intermolecular jaisa), constant density, B/A ≈ constant (latent heat jaisa), particle emission ↔ evaporation, drop splitting ↔ **fission** (Bohr–Wheeler).

WEIZSÄCKER FORMULA

$$B = a_vA - a_sA^{2/3} - a_c\frac{Z(Z-1)}{A^{1/3}} - a_a\frac{(A-2Z)^2}{A} \pm \delta \;;\; M = Zm_p + (A-Z)m_n - B/c^2$$

Term	Reason	≈ MeV
Volume a_vA	Saturation — har nucleon sirf neighbours se bound ⇒ $B \propto A$	15.5
Surface $-a_sA^{2/3}$	Surface nucleons ke kam neighbours; area $\propto A^{2/3}$	16.8
Coulomb $-a_cZ(Z-1)/A^{1/3}$	Proton–proton repulsion $\propto Z^2/R$	0.72
Asymmetry $-a_a(A-2Z)^2/A$	$N \neq Z \Rightarrow$ stability kam (Pauli)	23
Pairing ±δ	+ even-even, 0 odd-A, − odd-odd; $\delta \approx 34A^{-3/4}$	~12/√A

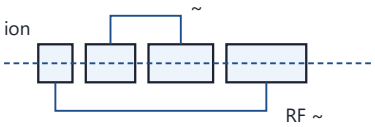
Success: B/A curve, fission, stable isobar. **Failure:** magic numbers, spin, magnetic moment, excited states.

2. Nuclear Shell Model

Atomic shells ki tarah nucleons average central potential mein independently move karte hue discrete **shells** bharte hain (p aur n alag, Pauli principle). **Magic numbers** 2, 8, 20, 28, 50, 82, 126 pe nuclei extra stable (high B/A, zyada isotopes, low n-absorption) — e.g. $_2He^4$, $_8O^{16}$, $_{20}Ca^{40}$, $_{82}Pb^{208}$ (doubly magic). Strong **spin–orbit coupling** (Mayer–Jensen) se $j = l \pm ½$ splitting magic numbers deta hai. **Success:** magic numbers, ground state spin–parity, magnetic moments, isomerism. **Failure:** fission, large quadrupole moments.

3. Particle Accelerators

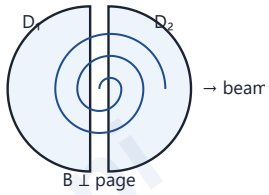
(a) Linear Accelerator: Coaxial **drift tubes**, alternate tubes RF source ke opposite terminals pe. Tube ke andar field zero (drift), gaps mein acceleration. Har tube mein time T/2 rakhne ke liye length badhti: $L_n = (1/2f)\sqrt{(2nqV/m)} \propto \sqrt{n}$; KE after n gaps = nqV.



Linac drift tubes

(b) Cyclotron: Do hollow **dees** strong uniform B mein, beech mein HF AC. Dee ke andar semicircle (field-free), gap mein acceleration; spiral path.

- $r = mv/qB$; semicircle time $\pi m/qB$ — v, r se independent.
- **Resonance:** $f = qB/2\pi m$; $E_{max} = q^2B^2R^2/2m$.
- **Limits:** relativistic mass increase ⇒ out of phase (p ~25 MeV max); electrons & neutrals nahi.



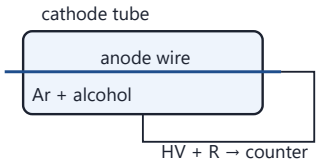
Cyclotron (top view)

NUMERICAL Proton, B = 1.5 T: $f = (1.6\times10^{-19}\times1.5)/(2\pi\times1.67\times10^{-27}) \approx$ **22.9 MHz**. R = 0.5 m ⇒ $E_{max} = (1.6\times10^{-19}\times1.5\times0.5)^2/(2\times1.67\times10^{-27}) = 4.31\times10^{-12}$ J ≈ **27 MeV**.

- **(c) Synchrocyclotron (FM cyclotron):** B constant, oscillator **frequency time ke saath ghatate** hain ($f = qB/2\pi m$, m badhta) ⇒ relativistic problem solved. Beam **pulsed**; protons ~700 MeV.
- **(d) Synchrotron:** Particles **fixed radius** ring (doughnut) mein; energy ke saath **B aur f dono** badhate hain ($r = p/qB$ constant). Ring ke around bending magnets + RF cavities. Proton synchrotron (GeV–TeV, LHC); electron synchrotron mein synchrotron radiation loss limit.
- **(e) Betatron (Kerst, electrons):** Doughnut chamber, fixed orbit r_0; acceleration **changing flux se induced E-field** (transformer, electron = secondary). Energy/rev = e dφ/dt. **Betatron (2:1) condition:** $B_{orbit} = ½\ B_{avg\ inside}$, i.e. $\varphi = 2\pi r_0^2 B_0$. AC ke first quarter cycle mein hi acceleration; ~100–300 MeV; X-rays, therapy.

4. Geiger–Müller Counter

Metal tube (cathode) + axial thin wire (anode, ~ +1000 V). Gas: argon + quenching agent (alcohol/halogen) at low pressure. Radiation gas ionize karti → electrons anode ki taraf accelerate → **avalanche (Townsend)** poore wire pe ⇒ large pulse (initial ionization se independent) → counter. Quencher continuous discharge rokta; **dead time** ~ 100–200 μs. Operating voltage = **plateau** region. β ~100% efficient, γ ~1%; particle type/energy nahi batata.



GM tube

5. Motion in Crossed Electric & Magnetic Fields

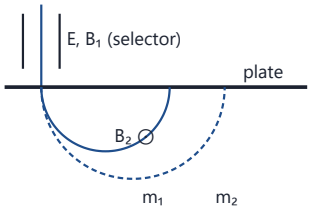
$F = q(E + v \times B)$, E ⊥ B. **Velocity selector:** $qE = qvB \Rightarrow v = E/B$ wale particles undeflected (Wien filter, Thomson e/m). Rest se start ⇒ **cycloidal** path, drift velocity $v_d = E/B$ along E × B (q, m se independent). Only B ⇒ circle ($r = mv/qB$); only E ⇒ parabola.

6. Mass Spectrographs

(a) Bainbridge: Ions slits se → **velocity selector** ($E \perp B_1$), sirf $v = E/B_1$ pass → chamber mein B_2 ⇒ semicircle → photographic plate.

$r = mv/qB_2 = mE/(qB_1B_2) \Rightarrow m = qB_1B_2r/E \propto r$

⇒ **linear mass scale**; isotopes: $\Delta m = qB_1B_2\Delta r/E$.



Bainbridge (schematic)

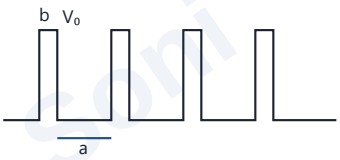
(b) Aston: Ion beam pehle **electric field** se ek taraf deflect, phir **magnetic field** se opposite taraf. Same e/m (different v) wale ions plate pe **ek point pe focus** — **velocity focusing**; different masses ⇒ alag lines, approx. linear scale. Aston ne isotopes (Ne-20, Ne-22) aur **whole number rule** diya; accuracy ~1 in 10^4. *Bainbridge = velocity selection (exactly linear scale); Aston = velocity focusing.*

UNIT IV Solid State Physics

1. Kronig–Penney Model (Qualitative)

Crystal ka periodic potential = 1-D periodic **rectangular wells** (barrier V_0, width b, period a+b). Schrödinger eq. + Bloch theorem se:

$$P\,(\sin \alpha a)/(\alpha a) + \cos \alpha a = \cos ka\,,\ P = mV_0ba/\hbar^2$$



K-P periodic potential

- cos ka sirf −1 se +1 ⇒ LHS jahan is range mein ho wahi E allowed ⇒ **allowed bands** + **forbidden gaps**.
- E badhe ⇒ band width badhe. P → ∞: discrete levels; P → 0: free electron.
- E–k mein gaps at k = nπ/a (Brillouin zone edges).

2. Effective Mass

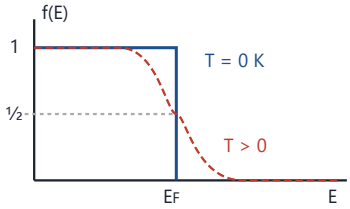
Crystal mein lattice ke internal force ka effect ek modified mass mein daalte hain taaki $F_{ext} = m^*a$: $m^* = \hbar^2 / (d^2E/dk^2)$.

Band bottom pe m* > 0; band top pe m* < 0 ⇒ **hole** (positive charge, positive mass) concept; inflection pe m* → ∞. Narrow band ⇒ heavy m*.

3. Fermi–Dirac Distribution

F-D FUNCTION

$$f(E) = \frac{1}{1 + e^{(E - E_F)/kT}}$$



f(E) vs E

- f(E) = level E ke occupied hone ki probability.
- T = 0: $E < E_F$ ⇒ f = 1, $E > E_F$ ⇒ f = 0.
- T > 0: $E = E_F$ pe **f = ½**. **Fermi level** = ½ occupancy level (0 K pe highest filled).
- $E - E_F \gg kT$ ⇒ Maxwell–Boltzmann.

4. Fermi Level: Intrinsic & Extrinsic Semiconductors

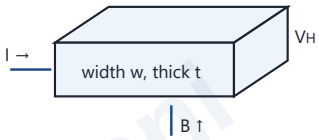
Type	Fermi level position
Intrinsic	$E_F = (E_c + E_v)/2 + (3kT/4)\ln(m_h^*/m_e^*)$ ⇒ $m_h^* = m_e^*$ ho to exactly **band gap ke middle**.
n-type (P, As)	0 K: $E_F = (E_c + E_d)/2$. Doping ↑ ⇒ E_c ki taraf; T ↑ ⇒ middle ki taraf girta. $E_c - E_F = kT \ln(N_c/N_D)$.
p-type (B, Al)	0 K: $E_F = (E_v + E_a)/2$. Doping ↑ ⇒ E_v ki taraf; T ↑ ⇒ middle ki taraf uthta. $E_F - E_v = kT \ln(N_v/N_A)$.

5. Zener, Tunnel, Photodiode & Solar Cell

Device	Principle	Characteristic / Use
Zener diode	Heavily doped, reverse bias. **Zener breakdown** (V_Z < ~5 V, field tunnelling) ya **avalanche** (higher V).	Breakdown pe V nearly constant ⇒ **voltage regulator**, reference.
Tunnel diode	Very heavy doping, depletion ~10 nm ⇒ quantum **tunnelling**.	I–V mein **negative resistance** (peak→valley). GHz oscillator, fast switch.
Photodiode	Reverse biased p-n/PIN; $h\nu \geq E_g$ ⇒ e–h pairs ⇒ reverse current ∝ intensity.	Very low dark current. Light detector, fiber receiver.
Solar cell	Large-area p-n, **no bias**; photovoltaic effect, junction field e–h separate karta.	V_{oc} ~0.5–0.6 V (Si); FF = $V_mI_m/V_{oc}I_{sc}$; η ~15–22%.

6. Hall Effect

Current-carrying slab ko transverse B mein rakho ⇒ I aur B dono ke ⊥ **Hall voltage V_H** (Lorentz force se charges ek side jama). Equilibrium: $eE_H = evB$, $J = nev \Rightarrow E_H = JB/ne$.



Hall setup

HALL RELATIONS

$$R_H = \frac{1}{ne} = \frac{V_H\, t}{I\, B} \;;\; \mu = \sigma R_H$$

R_H −ve ⇒ n-type, +ve ⇒ p-type. Uses: carrier type, n, mobility, Hall probe (B measure). **Numerical:** I = 10 mA, B = 0.5 T, t = 1 mm, V_H = 1 mV ⇒ $R_H = 2\times10^{-4}$ m³/C ⇒ $n = 1/eR_H = 3.1\times10^{22}$ m^{-3}.

7. Superconductivity

T_c se niche **zero resistance** (Onnes, Hg 4.2 K). $H_c(T) = H_c(0)[1 - (T/T_c)^2]$; H > H_c ya I > I_c ⇒ normal. Isotope effect $T_c \propto M^{-1/2}$. **Meissner effect:** T_c se niche cool karne par flux **expel**: B = μ_0(H + M) = 0 ⇒ M = −H, **χ = −1** (perfect diamagnetism) — levitation.

Type I (soft)	Type II (hard)
Single H_c, sharp transition, complete Meissner till H_c; low H_c (~0.1 T). Pure metals: Hg, Pb, Sn, Al.	H_{c1}, H_{c2}; between them **mixed/vortex state** (partial flux). High H_{c2} (tens of T). Alloys: NbTi, Nb_3Sn, YBCO — MRI magnets.

8. Dielectrics: Polarization, Complex Permittivity, Losses

Field mein dipoles induce/align: $P = \varepsilon_0(\varepsilon_r - 1)E$, $D = \varepsilon_0 E + P$.

- **Electronic:** electron cloud shift; $\alpha_e = 4\pi\varepsilon_0 R^3$; T-independent; ~$10^{15}$ Hz.
- **Ionic:** ions opposite shift (NaCl); ~10^{13} Hz.
- **Orientational:** permanent dipoles (H_2O); $\alpha_o = p^2/3kT$; ~10^{10} Hz.
- **Space charge:** interfaces pe charge; ~10^3 Hz.

COMPLEX PERMITTIVITY

$$\varepsilon^* = \varepsilon' - i\varepsilon'' \;;\; \tan\delta = \varepsilon''/\varepsilon' \;;\; \text{Power loss/volume} = \omega\varepsilon_0\varepsilon'' E^2_{rms}$$

AC mein dipoles field ke peeche lag karte (relaxation) ⇒ D aur E mein phase lag δ. ε′ = stored energy; ε″ = **loss factor** (heat). **Dielectric loss:** electrical energy → heat (dipole relaxation, leakage conduction, ionization); max at ωτ = 1. Good insulator: low tan δ (mica ~10^{-4}). Uses: capacitors, cables, microwave heating.

UNIT V Laser and Fiber Optics

1. Spontaneous & Stimulated Processes

Process	What happens	Rate
Absorption	E_1 atom photon hν = $E_2 - E_1$ absorb karke E_2 jaata	$B_{12}N_1\rho(\nu)$
Spontaneous emission	Excited atom khud se random time/direction/phase pe photon — incoherent	$A_{21}N_2$
Stimulated emission	Incident photon trigger ⇒ 2 identical photons (same ν, phase, direction, polarization) — coherent, amplification	$B_{21}N_2\rho(\nu)$

2. Einstein's A & B Coefficients, Transition Probabilities

Equilibrium: $B_{12}N_1\rho = A_{21}N_2 + B_{21}N_2\rho \Rightarrow \rho = A_{21}/(B_{12}(N_1/N_2) - B_{21})$. Boltzmann $N_1/N_2 = e^{h\nu/kT}$ daal kar Planck $\rho = (8\pi h\nu^3/c^3)/(e^{h\nu/kT} - 1)$ se compare:

EINSTEIN RELATIONS

$$B_{12} = B_{21} \;;\; A_{21}/B_{21} = 8\pi h\nu^3/c^3$$

Transition probabilities (per sec): spontaneous = A_{21} (= 1/τ), stimulated = $B_{21}\rho$, absorption = $B_{12}\rho$. Ratio $A_{21}/B_{21}\rho = e^{h\nu/kT} - 1 \gg 1$ at optical ν ⇒ normally spontaneous dominate; isliye population inversion + resonator chahiye.

3. Active Medium, Population Inversion, Pumping, Resonator

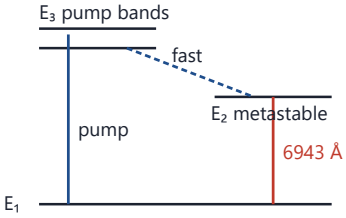
- **Active medium:** material jismein inversion aur lasing hoti (Cr^{3+} in ruby, Ne in He-Ne, CO_2).
- **Population inversion:** $N_2 > N_1$ (normally $N_2 < N_1$). **Metastable state** ($\tau \sim 10^{-3}$ s vs 10^{-8} s) zaroori; 2-level mein impossible, 3/4-level scheme use (4-level easier).
- **Pumping:** optical (flash lamp — ruby, Nd:YAG), electrical discharge (He-Ne, CO_2), chemical, current injection (diode laser).
- **Optical resonator:** do mirrors (100% + partial ~99%) ⇒ feedback, amplification, sirf axial photons survive; $L = n\lambda/2$ (modes). Types: plane-parallel, confocal.

4. Characteristics of Laser Beam

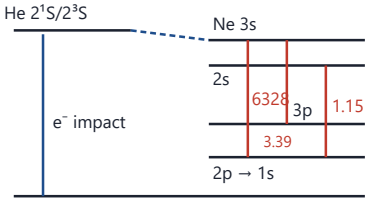
- **Coherence:** temporal ($l_c = c\tau_c$, He-Ne ~ km tak) & spatial.
- **Divergence:** $\theta = (r_2 - r_1)/(d_2 - d_1) \approx 1.22\lambda/D$, ~mrad.
- **Directionality:** nearly parallel beam.
- **Monochromatic** ($\Delta\lambda \sim 10^{-3}$ nm) & **high intensity**.

5. Ruby, Nd:YAG, He-Ne & CO_2 Lasers

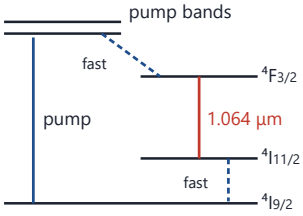
	Ruby	Nd:YAG	He-Ne	CO_2
Type	Solid, 3-level	Solid, 4-level	Atomic gas, 4-level	Molecular gas, 4-level
Active	Cr^{3+} (0.05%) in Al_2O_3	Nd^{3+} in $Y_3Al_5O_{12}$	Ne (He:Ne ≈ 10:1)	$CO_2 + N_2 + He$
Pump	Xenon flash lamp	Kr/Xe lamp, diode	Electric discharge	Electric discharge
λ	6943 Å	1.064 µm	6328 Å	10.6 µm (9.6 µm)
Output	Pulsed, η ~1%	Pulsed/CW, high power	CW, mW	CW, kW, η ~30–40%



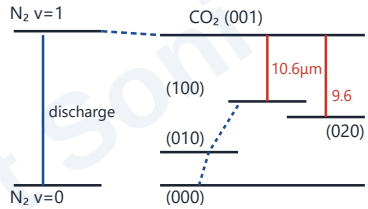
Ruby (3-level)



He-Ne energy levels



Nd:YAG (4-level): lower laser level $^4I_{11/2}$ ground se upar



CO_2: $N_2 \rightarrow CO_2(001)$ resonant transfer; (001)→(100) = 10.6 µm

- **Ruby:** Flash lamp Cr^{3+} ko $E_1 \rightarrow E_3$ (green/blue absorb) → fast non-radiative E_2 (metastable, τ ~3 ms) → $E_2 \rightarrow E_1$ stimulated, 6943 Å. Lower level = ground ⇒ >50% atoms excite karne padte ⇒ pulsed, low η.
- **Nd:YAG:** ground $^4I_{9/2}$ → pump bands → fast to **$^4F_{3/2}$** (metastable) → lasing to **$^4I_{11/2}$** (1.064 µm) → fast to ground. Lower level ground se upar ⇒ easy inversion (4-level).
- **He-Ne:** Discharge He ko 2^1S, 2^3S (metastable) pe; collision se Ne 3s, 2s ko resonant transfer (He 20.61 eV ≈ Ne 20.66 eV). 3s→2p = 6328 Å, 2s→2p = 1.15 µm, 3s→3p = 3.39 µm; 2p→1s spontaneous, 1s→ground tube wall collision se.
- **CO_2:** Vibrational modes: symmetric (100), bending (010), asymmetric (001). Discharge N_2 (v = 1) excite → resonant transfer to CO_2 (001). Lasing **(001)→(100) = 10.6 µm**, (001)→(020) = 9.6 µm. He lower level empty + cooling karta.

6. Optical Fiber Basics & Types

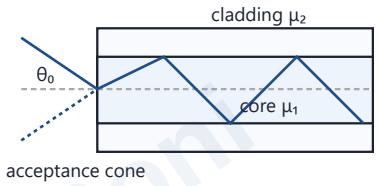
Core (μ_1) + **cladding** ($\mu_2 < \mu_1$) + jacket. Principle: **TIR** at core-cladding, $\sin\theta_c = \mu_2/\mu_1$.

Type	Core dia.	Features
Single-mode step index	~8–10 µm	Ek mode, no intermodal dispersion, laser source, long haul
Multimode step index	50–200 µm	Zig-zag rays, high intermodal dispersion, LED, short links
Multimode graded index	50–85 µm	µ centre pe max (parabolic); curved paths, low dispersion

7. Acceptance Angle, Cone, NA, V-number (Ray Theory)

Ray axis se θ_0 pe enter, core mein θ_r, interface pe $(90° - \theta_r) \geq \theta_c$ chahiye (TIR, zig-zag propagation).

- $\mu_0 \sin\theta_0 = \mu_1 \sin\theta_r = \mu_1 \cos\theta_c$ (limit).
- $\cos\theta_c = \sqrt{(1 - \mu_2^2/\mu_1^2)} \Rightarrow \mu_0 \sin\theta_{max} = \sqrt{(\mu_1^2 - \mu_2^2)}$.
- **Acceptance angle** θ_{max}; isse axis ke around ghumao ⇒ **acceptance cone** (full $2\theta_{max}$).



Step index: TIR zig-zag

NA & V-NUMBER

$$NA = \mu_0 \sin\theta_{max} = \sqrt{(\mu_1^2 - \mu_2^2)} \approx \mu_1\sqrt{(2\Delta)},\ \Delta = (\mu_1 - \mu_2)/\mu_1\ ;\ V = (2\pi a/\lambda)\cdot NA$$

Modes ≈ $V^2/2$ (step), $V^2/4$ (graded). **V < 2.405 ⇒ single mode. Numerical:** μ_1 = 1.50, μ_2 = 1.47 ⇒ NA = √0.0891 = **0.298**, θ_{max} = **17.3°**, Δ = 0.02. a = 25 μm, λ = 1.3 μm ⇒ V ≈ 36, modes ≈ 648.

8. Pulse Dispersion, Attenuation & Losses

Dispersion = pulse ka broadening ⇒ overlap ⇒ bit rate limit. **Intermodal** (multimode, different path lengths): step index Δt ≈ $L\mu_1\Delta/c$; graded ≈ $L\mu_1\Delta^2/8c$. **Intramodal/chromatic:** material (μ depends on λ) + waveguide dispersion.

Attenuation: $\alpha\ (dB/km) = (10/L)\log_{10}(P_{in}/P_{out})$; silica min ~0.2 dB/km at 1.55 μm (windows 0.85, 1.3, 1.55 μm). **Losses:** absorption (intrinsic UV/IR, impurities OH^-, Fe, Cu), **Rayleigh scattering** ∝ $1/\lambda^4$, macro/micro **bending**, splice/connector/coupling losses. **Numerical:** 1 mW, 10 km, 0.5 dB/km ⇒ 5 dB ⇒ $P_{out} = 10^{-0.5}$ mW = **0.316 mW**.

9. Applications of Lasers & Optical Fibers

Lasers: cutting, welding, drilling (CO_2, Nd:YAG); eye surgery (LASIK, retina), bloodless surgery, cancer therapy; fiber communication source, holography, barcode, CD/DVD, printers; LIDAR, range finding, spectroscopy, isotope separation.

Optical fibers: telecom & internet (high bandwidth, low loss, no EMI, secure, light); medical endoscopy, laser delivery; sensors (temperature, pressure, strain); LAN, cable TV, defence/aircraft, decorative lighting.

Quick Revision — Must-know Formulas

Unit	Formulas
I	$v_g = v_p - \lambda dv_p/d\lambda$; $v_p v_g = c^2$ · $\Delta x\cdot\Delta p \geq \hbar/2$ · $\Delta\lambda = (h/m_0c)(1 - \cos\varphi)$ · $\hat{p} = -i\hbar\partial/\partial x$, $\hat{E} = i\hbar\partial/\partial t$ · $E_n = n^2h^2/8mL^2$, $\psi_n = \sqrt{(2/L)}\sin(n\pi x/L)$
II	$\beta = \lambda D/d$ · $2\mu t\cos r = n\lambda$ (dark, reflected) · $\beta = \lambda/2\mu\theta$ · $D_n^2 = 4n\lambda R$ · $2d = N\lambda$, $\Delta\lambda = \lambda^2/2d$ · $(a+b)\sin\theta = n\lambda$ · RP = nN, $t\ d\mu/d\lambda\cdot\mu = \tan i_p$ · $t_{QWP} = \lambda/4(\mu_o - \mu_e)$
III	SEMF (5 terms) · $f = qB/2\pi m$, $E_{max} = q^2B^2R^2/2m$ · $L_n \propto \sqrt{n}\cdot B_{orbit} = ½B_{avg}$ · $v = E/B$ · $m = qB_1B_2r/E$
IV	$m^* = \hbar^2/(d^2E/dk^2)$ · $f(E) = 1/(1 + e^{(E-E_F)/kT})$ · $R_H = 1/ne = V_Ht/IB$ · $H_c(T) = H_c(0)[1 - (T/T_c)^2]$ · $\varepsilon^* = \varepsilon' - i\varepsilon''$, $\tan\delta = \varepsilon''/\varepsilon'$
V	$A/B = 8\pi h\nu^3/c^3$, $B_{12} = B_{21}$ · $NA = \sqrt{(\mu_1^2 - \mu_2^2)}$ · $V = 2\pi a\cdot NA/\lambda$ · $\Delta t = L\mu_1\Delta/c$ · $\alpha = (10/L)\log(P_{in}/P_{out})$

COMMON MISTAKE Newton's rings mein centre **dark** (reflected) — bright nahi. Ruby 3-level hai, baaki teen (Nd:YAG, He-Ne, CO_2) 4-level. He-Ne mein lasing **Ne** atoms deti hai, He sirf energy transfer karta. Compton shift incident λ pe depend nahi karta.

YAAD RAKHO Ruby 6943 Å | He-Ne 6328 Å | Nd:YAG 1.064 μm | CO_2 10.6 μm · NA = $\sqrt{(\mu_1^2-\mu_2^2)}$, single mode V < 2.405 · $A/B = 8\pi h\nu^3/c^3$ · Compton h/m_0c = 0.0243 Å · $E_n = n^2h^2/8mL^2$.

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